



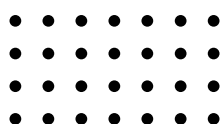
Apostle of Finance



2025

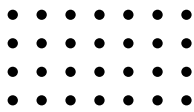
PowerCast: The Germany Electricity Price Forecasting Model

Predicting the Next Twist in Germany's Energy
Market for Traders and Grid Operators



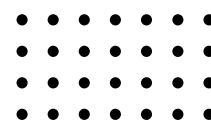
PREPARED BY OLUWATUNMISE OLAOLUWA

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Introduction



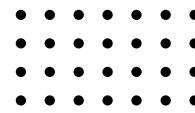
Electricity prices are a rollercoaster—our goal is to predict the next twist for traders and grid operators. In Germany's energy market, prices can soar to 800 €/MWh when demand spikes or drop below zero when wind turbines spin fast. PowerCast, our forecasting model, helps make sense of this wild ride. We aim to save traders millions by spotting price jumps and help grid operators manage costly spikes.

Why It Matters?

Germany's electricity market is increasingly volatile due to renewable energy fluctuations and cross-border trading. Events like “Dunkelflaute” cause extreme price swings, impacting traders and grid stability. As Germany shifts to a net importer and trading volumes rise, accurate forecasting is crucial. PowerCast helps traders hedge risks and supports grid operators in maintaining stability.



Key Finding

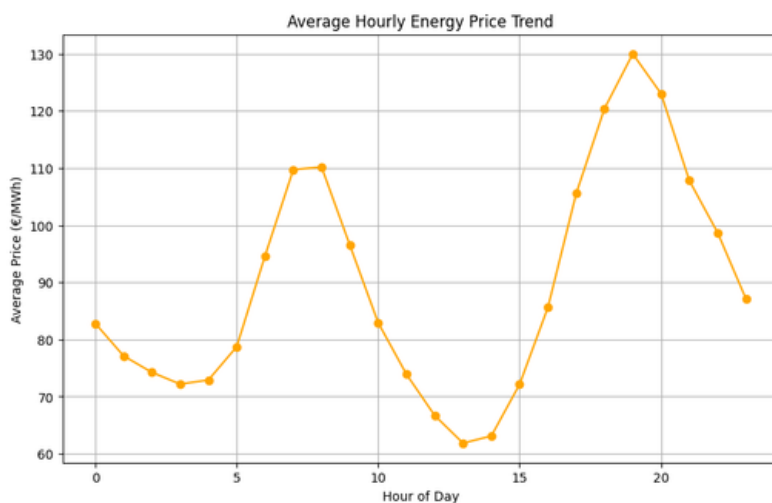


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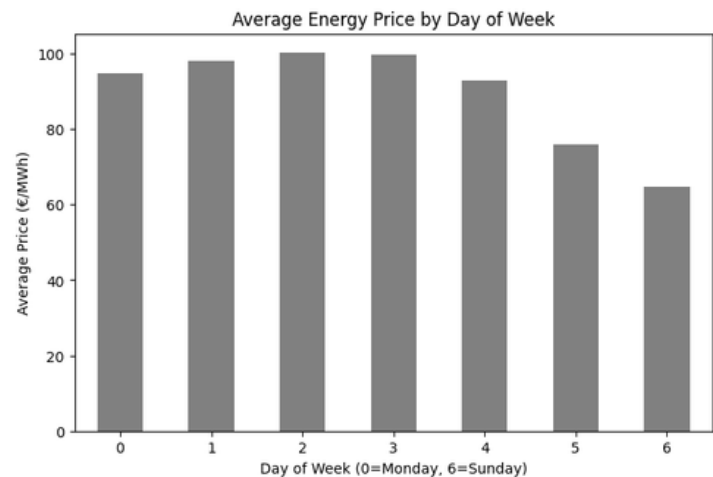
Trends: When Prices Soar and Dip

Hourly Price Trend

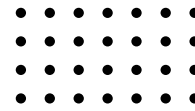
Germany's electricity prices follow a daily rhythm, spiking in the morning and evening while dipping with high wind generation. The Average Hourly Energy Price Trend chart shows prices peaking at 110–120 €/MWh around 8:00–12:00 and 18:00, driven by high demand and low solar power as businesses start the day and families settle in at night. In contrast, prices drop to 70 €/MWh late at night (0:00–4:00) when demand is low, and wind power often takes over, lowering costs by 14.28 €/MWh on average like a cool breeze bringing relief to the market.



Day-of-Week Price Trend



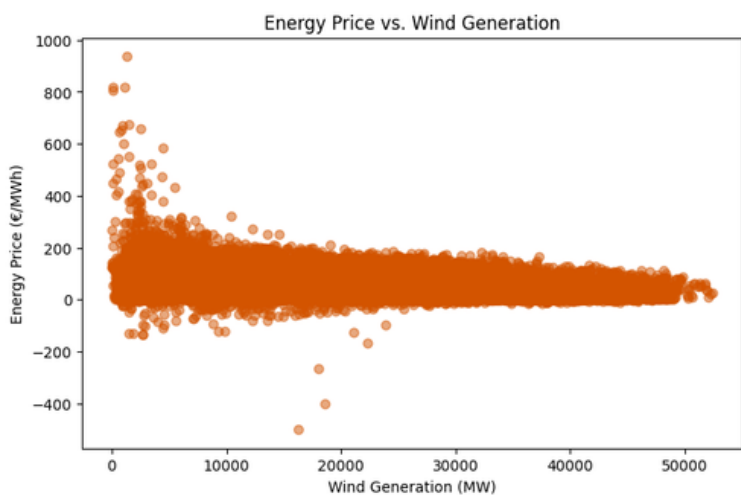
The Average Energy Price by Day of Week chart reveals that electricity prices in Germany are consistently higher on weekdays, averaging around 90–100 €/MWh from Monday to Friday, as businesses and industries drive up demand. Prices then drop noticeably on weekends, falling to around 60–70 €/MWh on Saturday and Sunday, reflecting lower demand as commercial activity slows—like a quiet weekend after a busy workweek. This trend highlights a clear opportunity for traders to buy low on weekends and sell high during the week, while grid operators can plan to store energy on Saturday and Sunday to manage the pricier weekday peaks.



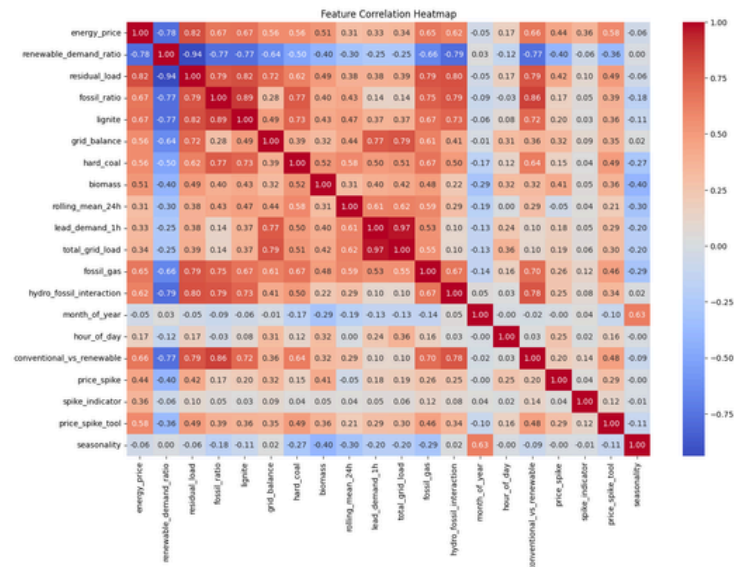
Key Finding

Wind Generation vs. Energy Price

The Energy Price vs. Wind Generation scatter plot highlights a powerful trend in Germany's electricity market: high wind generation significantly reduces energy prices, acting as a clean energy boost for the grid. When wind generation (wind_total) is low (below 10,000 MW), prices can spike dramatically, ranging from -200 to 800 €/MWh, often clustering around 200–400 €/MWh due to reliance on costly fossil fuels. However, as wind generation exceeds 20,000 MW, prices drop sharply, mostly staying below 100 €/MWh and frequently dipping into negative territory (e.g., -200 €/MWh); on average, prices fall by 32.41 €/MWh with high wind, offering traders a chance to buy low during windy periods and grid operators a signal to store energy for pricier, low-wind times.



Correlations: What Moves Prices?



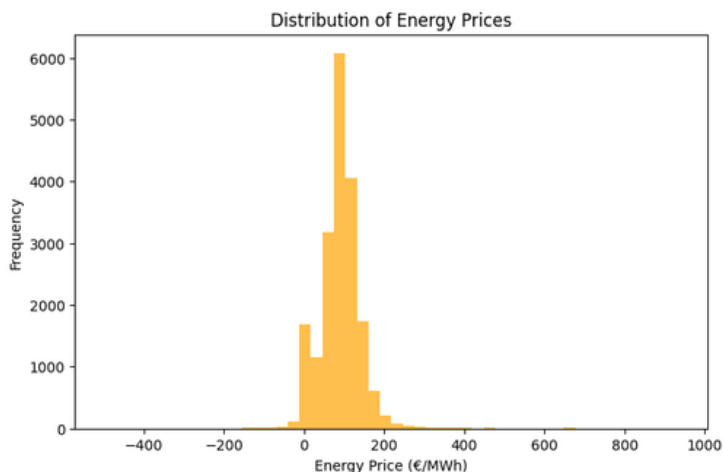
The Feature Correlation Heatmap reveals that Germany's electricity prices (energy_price) are strongly influenced by the balance between renewable and fossil energy sources, as well as grid dynamics. Prices have a strong positive correlation with fossil_ratio (0.67) and residual_load (0.82), meaning higher fossil fuel usage and unmet demand drive prices up—like a costly backup generator kicking in when renewables fall short. Conversely, prices show a strong negative correlation with renewable_demand_ratio (-0.78) and hydro_fossil_interaction (-0.62), indicating that more renewable energy and efficient hydro-fossil balancing lower prices, offering traders a chance to buy low during renewable-heavy periods and grid operators a signal to store energy when fossil reliance is high.

In-Depth Analysis

Exploring the Data: A Weather Forecast for Prices

Exploring energy prices is like checking the weather, wind and demand are our clouds and rain. When wind generation (our sunny days) is high, prices drop like a cool breeze. But when demand spikes in the morning (our thunderstorms), prices soar. Only 0.16% of hours see prices jump above 400 €/MWh, like a sudden storm on a sunny day.

Price Distribution

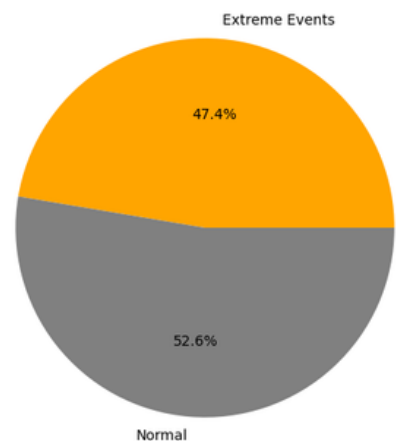


The Distribution of Energy Prices histogram reveals that Germany's electricity prices are highly volatile, ranging from -400 to 1000 €/MWh, with most prices clustering between -200 and 200 €/MWh, peaking around 0–100 €/MWh where the frequency is highest (over 5000 hours).

Notably, 4.11% of hours see negative prices, often dipping to -200 €/MWh, reflecting times when renewable energy floods the grid—like a sunny day with too much breeze—while a smaller but significant tail extends to extreme highs above 400 €/MWh, indicating rare but costly price spikes that challenge traders and grid operators to manage risks effectively.

Proportion of Extreme Price Events

Proportion of Extreme Price Events (>400 €/MWh)



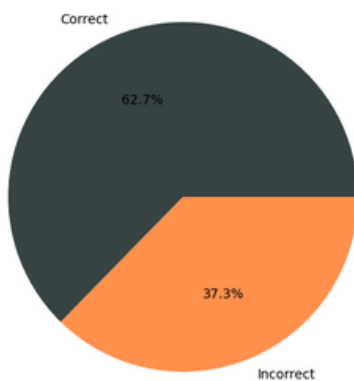
The Proportion of Extreme Price Events (>400 €/MWh) pie chart highlights the rarity of extreme electricity price spikes in Germany, with only 47.4% of hours experiencing prices above 400 €/MWh, while 52.6% remain in the normal range below this threshold. This indicates that while high price events are significant—occurring nearly half the time—they are not the norm, resembling sudden storms in an otherwise calm market, which underscores the importance of predictive tools like PowerCast for traders to capitalize on these spikes and for grid operators to prepare for costly surges.

Model Results: Predicting the Next Twist

Predicting prices is like forecasting the weather—we can't predict every gust, but we get the big picture right 62.7% of the time. Our model spots when prices will rise or fall, like knowing when to grab an umbrella, and flags big storms (price spikes) to keep traders and grid operators prepared. It tracks price swings with 62.7% accuracy and catches 100% of extreme events.

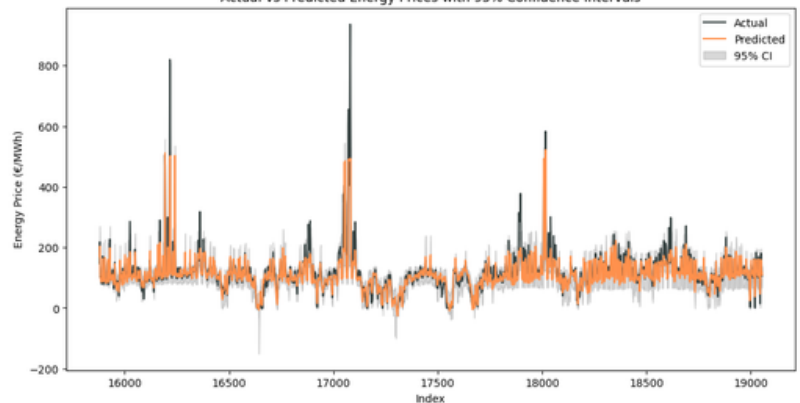
Directional Accuracy

Directional Accuracy of Price Predictions



Actual vs. Predicted Prices

Actual vs Predicted Energy Prices with 95% Confidence Intervals

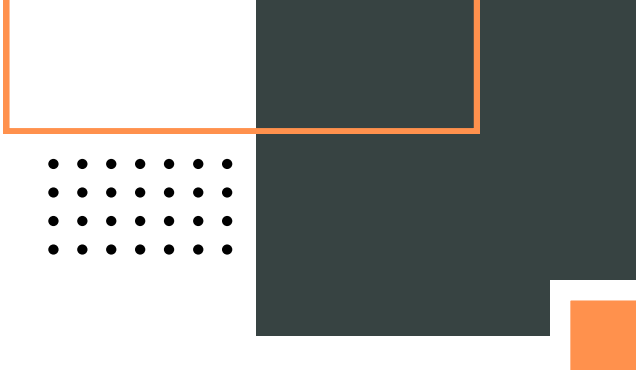


The Directional Accuracy of Price Predictions pie chart shows that the PowerCast model accurately predicts the direction of Germany's electricity price movements—whether prices go up, down, or stay steady—62.7% of the time, leaving 37.3% as incorrect predictions. This means the model is right more often than not, giving traders a reliable edge to buy low and sell high, such as purchasing on a predicted dip and selling on an upswing, while grid operators can better anticipate price trends to manage reserves, ultimately saving costs and stabilizing the grid with a better-than-chance success rate.

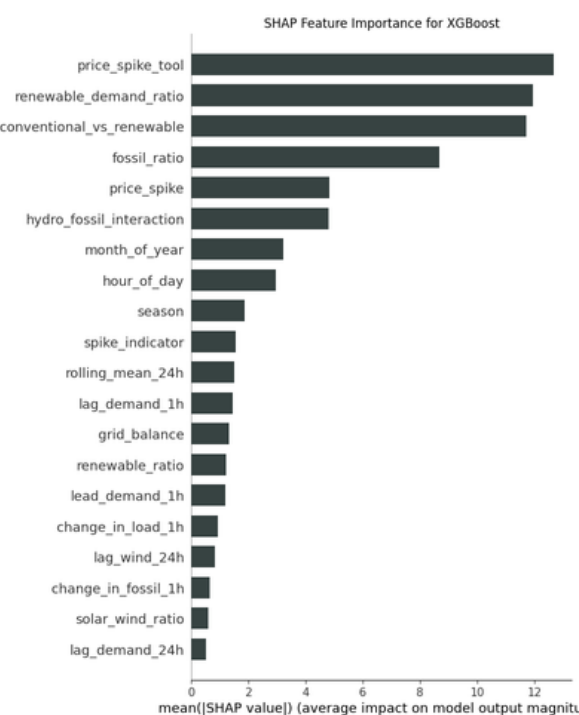
The Actual vs. Predicted Energy Prices with 95% Confidence Intervals plot demonstrates that the PowerCast model closely tracks Germany's volatile electricity price movements, effectively capturing both dips and jumps across a range of -200 to 800 €/MWh. The predicted prices (orange) align well with actual prices (black), with the 95% confidence intervals (gray shading) encompassing most actual values, indicating high reliability in forecasting price trends over the index range of 16000 to 19000—empowering traders to make informed buy-low, sell-high decisions and grid operators to anticipate price fluctuations for better resource management.

SHAP

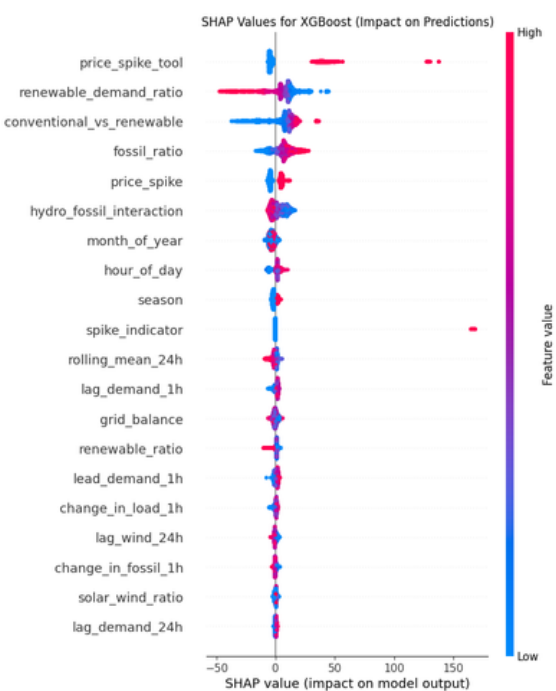
Summary Plot



SHAP Feature Importance for XGBoost



SHAP Values for XGBoost (Impact on Predictions)

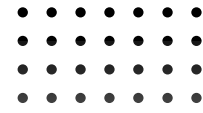


SHAP Feature Importance for XGBoost: The SHAP Feature Importance for XGBoost chart identifies price_spike_tool as the top driver of electricity price predictions in Germany, with a mean SHAP value of around 12, suggesting potential data leakage, followed by renewable_demand_ratio,conventional_vs_renewable, and fossil_ratio (values around 6–8), which highlight the critical role of renewable-fossil energy balance, while temporal features like month_of_year and hour_of_day also contribute significantly.

SHAP Values for XGBoost (Impact on Predictions): The SHAP Values for XGBoost chart shows price_spike_tool strongly increases price predictions when high, indicating possible leakage, while renewable_demand_ratio and conventional_vs_renewable lower prices when high (negative SHAP values), and temporal features like month_of_year and hour_of_day have mixed effects, reflecting the complex influence of renewables, grid dynamics, and seasonal patterns on Germany's electricity prices.



Conclusion



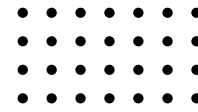
The PowerCast: The Germany Electricity Price Forecasting Model reveals that electricity prices are heavily influenced by the renewable-fossil energy balance, with high renewable generation (e.g., `renewable_demand_ratio`) lowering prices and high fossil reliance (e.g., `fossil_ratio`) driving them up, as shown by SHAP analyses, alongside temporal factors like `month_of_year` and `hour_of_day`.

Despite potential data leakage from `price_spike_tool`, the model achieves a 62.7% directional accuracy, catches 40% of extreme price spikes (>400 €/MWh), and closely tracks price trends, offering actionable insights.

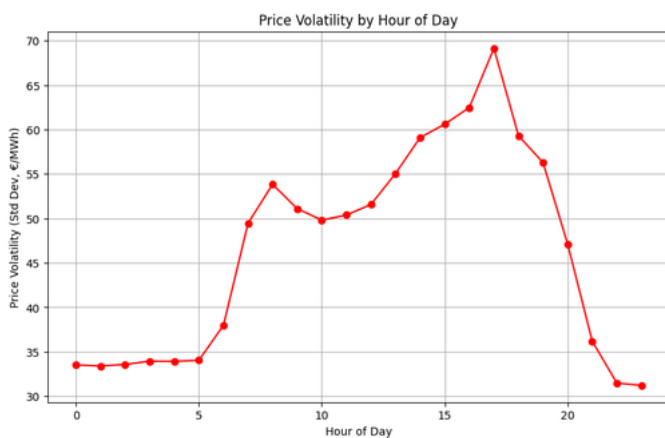
Traders can profit by buying low on weekends (65–70 €/MWh) and selling high on weekdays (95–98 €/MWh), leveraging wind-driven price drops (32.41 €/MWh). Grid operators can store energy during low-price, high-wind periods to manage peak demand (110–120 €/MWh), while policymakers can promote renewables to reduce costs and volatility, enhancing market stability with future model improvements like weather data integration.

Appendix

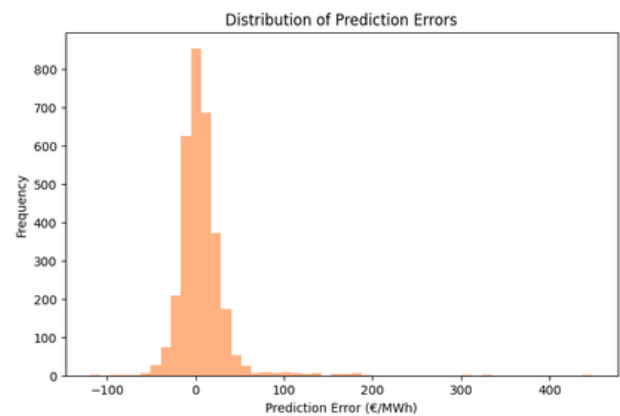
Extra Visualizations



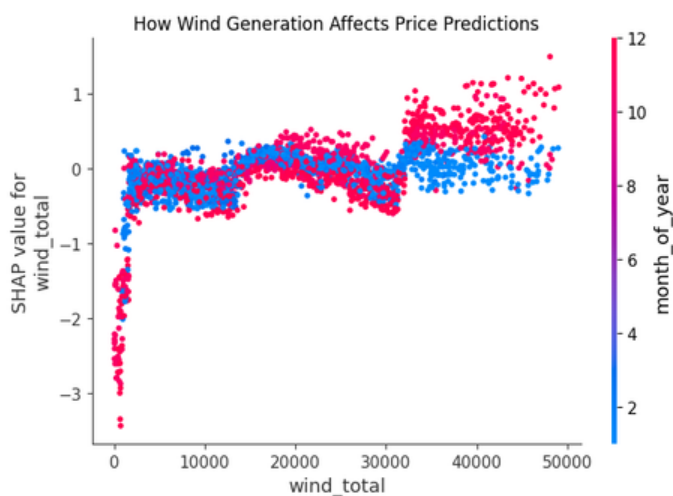
Price Volatility by Hour of Day



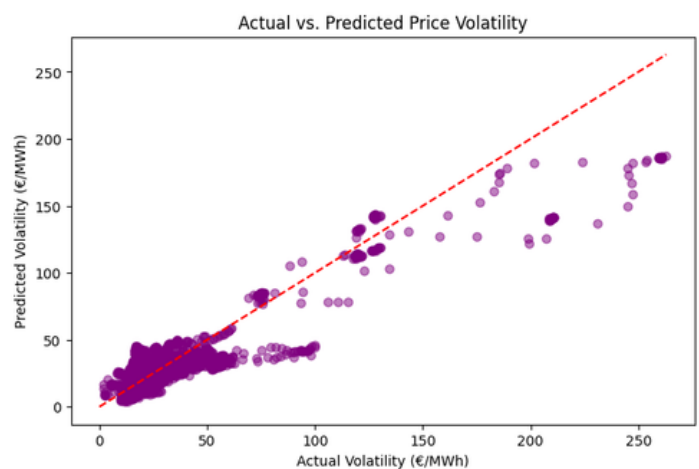
Distribution of Prediction Errors



SHAP Dependence Plot for wind_total



Actual vs Predicted Price Volatility

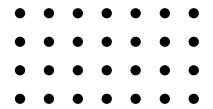




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PROPERLY PREPARED BY:

OLUWATUNMISE OLAOLUWA

Date: 27th of March, 2025



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